

# The shallow chamber after glaucoma surgery

Part 2 | Start with pressure and chamber geometry

A shallow chamber is a shared sign, not a single mechanism.

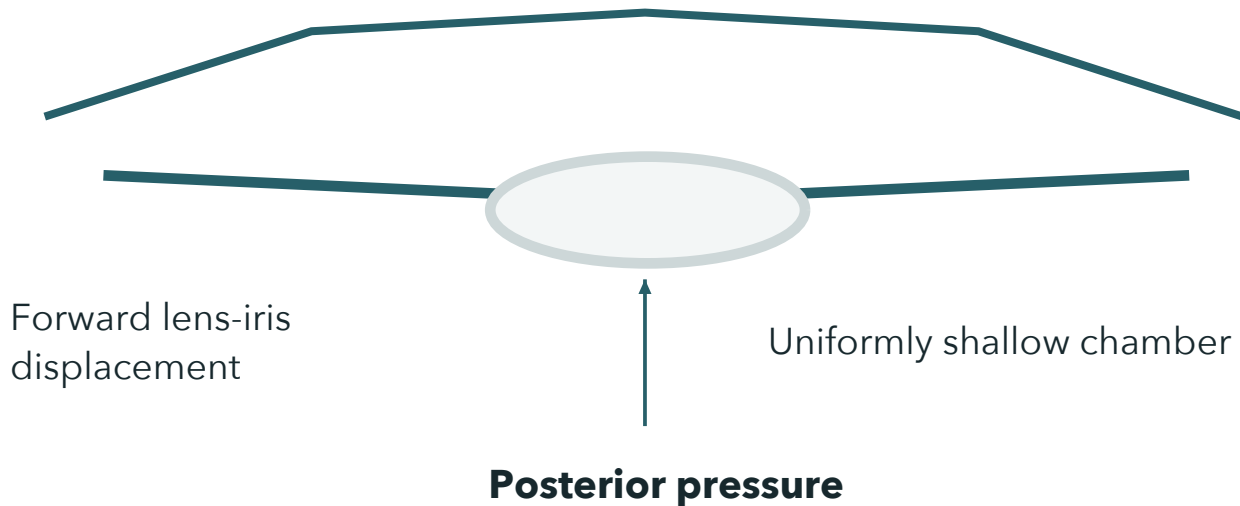
| Pressure | Depth pattern                   | Consider                       |
|----------|---------------------------------|--------------------------------|
| Low      | Shallow chamber                 | Leak / overdrainage            |
| Low      | Choroidal elevation             | Serous effusion                |
| High     | Peripheral > central shallowing | Pupil block                    |
| High     | Uniform shallowing; patent PI   | Misdirection / posterior cause |

Original conceptual map: depth pattern and IOP narrow the mechanism.

Check the wound and bleb, iridotomy patency, central versus peripheral depth and the posterior segment. Low-pressure overdrainage and high-pressure misdirection require different responses. Exclude hemorrhage and other posterior pathology when examination is limited. A patent iridotomy makes pure pupillary block less likely but does not itself establish misdirection.

# Uniform shallowing with a patent iridotomy

A posterior pressure-compartment problem can persist after pupil block is relieved.



Original schematic: pressure behind the lens-iris diaphragm shifts it forward.

Aqueous misdirection is a diagnosis after appropriate exclusions. Specialist treatment may include cycloplegia, aqueous suppression and selected interventions that establish a durable anterior-posterior communication. Miotics can aggravate forward displacement. Repeatedly adding anterior irrigation during suspected intraoperative infusion misdirection may worsen the problem rather than create lasting depth.

## Low pressure points towards a different branch

Examine drainage and choroidal anatomy together.

| Finding                               | Mechanism to assess                       |
|---------------------------------------|-------------------------------------------|
| Seidel-positive bleb or wound         | External leakage                          |
| Large filtration with no visible leak | Excessive outflow                         |
| Choroidal elevations at low IOP       | Serous effusion                           |
| Inflammation with low IOP             | Reduced aqueous production may contribute |

Conservative or surgical treatment depends on severity, chamber depth, corneal-lens contact and posterior complications. Control the cause of hypotony rather than pursuing a chamber-depth number alone. Large appositional choroidal detachments or persistent structural compromise require specialist reassessment of drainage and timing.

# Hypotony can damage vision even in a formed eye

Pressure, retinal folds and function are separate measurements.

| Observation                        | Interpretation                                   |
|------------------------------------|--------------------------------------------------|
| Chorioretinal folds                | A low-pressure structural consequence            |
| Disc swelling or vessel tortuosity | Can accompany altered globe geometry             |
| Poor acuity despite a clear cornea | Macular anatomy needs review                     |
| IOP below previous baseline        | Clinical harm depends on the eye, not one cutoff |

OCT can reveal subtle macular folds and document recovery. Identify leakage, overfiltration, inflammation or another source of low pressure. A chamber that appears reasonably deep does not exclude hypotony maculopathy. Persistent untreated folds may limit recovery even when pressure is later restored.

# Exclude suprachoroidal hemorrhage when the pattern changes

Sudden pain, loss of depth or red reflex demands immediate reassessment.

- Stop unsafe continuation and stabilize the eye.  
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- Assess pressure, red reflex and posterior findings.  
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- Use appropriate imaging when the fundus cannot be assessed.  
↓
- Coordinate definitive management from extent and evolving anatomy.

Hemorrhage and serous effusion are not interchangeable. Avoid assuming that every firm shallow eye is misdirection. Drainage timing and technique depend on liquefaction, extent and clinical threat. Antithrombotic decisions require individualized coordination; a blanket instruction to stop every anticoagulant ignores systemic risk.